KISAN INJECTION 1MG / 1ML (1ML AMP)
(Benzyl Alcohol Free)

WARNING - INTRAVENOUS USE
Severe reactions, including fatalities, have occurred during and immediately after the parenteral administration of KISAN INJECTION (Phytomenadione). Typically these severe reactions have resembled hypersensitivity or anaphylaxis, including shock and cardiac and/or respiratory arrest. Some patients have exhibited these severe reactions on receiving KISAN INJECTION for the first time. The majority of these reported events occurred following intravenous administration, even when precautions have been taken to dilute the KISAN INJECTION and to avoid rapid infusion. Therefore, the INTRAVENOUS route should be restricted to those situations where another route is not feasible and the increased risk involved is considered justified. When slow i.v injection is given, the rate should not exceed 1mg/min.

DESCRIPTION: Kisan Injection 1 mg / ml (1 ml amp) is a clear yellow solution.

COMPOSITION: Each 1 ml ampoule contains Vitamin K1 (Phytomenadione) 1 mg.
Phenol : 0.45% w/v Cremaphor EL : 5% w/v

PHARMACODYNAMICS: Class: (Vitamin K1).
Site & Mode of Action: As a component of an enzyme system, Vitamin K1 promotes the formation in the liver of coagulation factors II (prothrombin), VII, IX and X. Anticoagulants of the coumarin and indandione series cause a reversible displacement of Vitamin K1 from this enzyme system, thereby inhibiting the synthesis of these factors. Since this is a competitive displacement, Vitamin K1 is a specific antagonist for warfarin and similar anticoagulants. It is not capable, however, of terminating the action of heparin; for this purpose a salt of protamine should be used.

PHARMACOKINETICS: Systemic availability after i.m. administration is about 50%. The primary distribution compartment corresponds to the plasma volume. In blood plasma, 90% of Vitamin K1 is bound to lipoprotein (VLDL portion). Vitamin K1 plasma concentration is normally between 0.4 and 1.2 mcg./L. An i.m. dose of 10 mg Vitamin K1 produces plasma concentrations of 10-20 mcg/L.
The elimination half-life in plasma is 1.5 – 3 hours. After metabolic degradation Vitamin K1 is bound to glucuronic acid and excreted in the bile and urine.
Less than 10% of the drug is excreted unchanged in the urine. Vitamin K1 has an active metabolite, vitamin K1-2, 3 epoxide, which is reconverted into Vitamin K1.

INDICATIONS: Haemorrhage, antidote to anticoagulant drugs of the coumarin type, hypoprothrombinemia. Haemorrhage in the newborn: prophylaxis and therapy.

RECOMMENDED DOSAGE: When given IM, Vitamin K1 injection should be injected in the same doses as when given orally up to 5mg. The delay before the onset of action is approximately the same by both routes.
If surgical intervention should be necessary in a patient receiving anticoagulants of the coumarin or indandione series, their anticoagulant action, may if necessary be counteracted by means of Vitamin K1.
If there is recurrence of thrombosis while Vitamin K1 is being used, IV administration of heparin is recommended as a first measure.
Prophylaxis of Haemorrhagic Disease of the Newborn: The American Academy of Paediatrics recommends that Vitamin K1 be given to the newborn. A single intramuscular dose 0.5 to 1 mg within one hour of birth is recommended.
Treatment of Haemorrhagic Disease of the Newborn: Empiric administration of Vitamin K1 should not replace proper laboratory evaluation of the coagulation mechanism. A prompt response (shortening of the prothrombin time in 2 to 4 hours following administration of Vitamin K1 is usually diagnostic of haemorrhagic disease of the newborn, and failure to respond indicates another diagnosis or coagulation disorder.
Kisan 1 mg should be given either subcutaneously or intramuscularly. Higher doses may be necessary if the mother has been receiving oral anticoagulants. Whole blood or component therapy may be indicated if bleeding is excessive. This therapy, however, does not correct the underlying disorder and Kisan should be given concurrently.
In the treatment of vitamin K deficiency bleeding in neonates, phytomenadione may be given in a dose of 1mg intravenously, subcutaneously, or intramuscularly: further dose may be given if necessary. As a prophylactic measure, a single dose of 0.5 to 1mg may be given intramuscularly to the newborn infant, or 2mg orally followed by a second dose of 2mg after 4 to 7 days.

ROUTE OF ADMINISTRATION: For i.m, s.c., slow i.v. inj. or orally

CONTRAINDICATIONS: Vitamin K1 ampoules should not be used for patients with pronounced allergic diathesis or with a known history of a severe hypersensitivity reaction to agents containing Cremophor® EL or its derivatives such as polyoxyethylated castor oil.
The dosage in neonates of Vitamin K1 parenteral should not exceed 5 mg in the first few days of life because of the immaturity of the hepatic enzyme systems. Only 1 mg ampoules may be used in these patients.

WARNING & PRECAUTIONS: Vitamin K1 should be considered as adjunctive therapy to blood transfusions for severe haemorrhage due to anticoagulant therapy; it is not effective when heparin-like compounds have been used for anticoagulant therapy; minimal doses should be used to offset refractoriness to coumarin-like anticoagulants if long term anticoagulant therapy is intended.
Administration: In severe bleeding, or situations where other routes are not feasible, Vitamin K1 may be given by very slow intravenous injection, at a rate not exceeding 1 mg per minute.

INTERACTION WITH OTHER MEDICATIONS:

a) Concurrent use of anticoagulants, coumarin or indandione-derivative with Vitamin K may decrease the effects of these anticoagulants as a result of increased hepatic synthesis of pro-coagulant factors.
b) Requirements for Vitamin K may be increased in patients receiving broad-spectrum antibiotics, Quinidine, Quinine, high doses of salicylates or antibacterial sulphonamides.

USE IN PREGNANCY & LACTATION: Use in pregnancy: There is no specific evidence regarding the safety of Vitamin K1 in pregnancy and no reproductive studies have been performed in animals. Therefore, as with most drugs, administration during pregnancy should only occur if the benefits outweigh the risks.

SIDE EFFECTS: Intravenous injection of Vitamin K1 may, in rare cases, cause severe, anaphylactoid reactions. Should an anaphylactoid reaction occur, the usual measures must be taken (eg. administration of adrenaline and supportive measures as required). Intramuscular administration may cause local pain, sometimes with erythema at the injection site. There may also be tenderness.

SYMPTOMS AND TREATMENT OF OVERDOSE: Overdosage: Hypervitaminosis of Vitamin K1 has not been reported.

STORAGE CONDITIONS: Store below 30ºC. Protect from light. KEEP OUT OF REACH OF CHILDREN. *JAUHKAN DARIPADA KANAK-KANAK*. DO NOT ALLOW TO FREEZE. IF SEPARATION HAS OCCURRED OR IF OIL DROPLETS HAVE APPEARED, THE INJECTION SHOULD NOT BE USED.

PACK SIZE: Packs of 10 ampoules x 1 ml

SHELF LIFE: Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER: DUOPHARMA (M) SDN BHD
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